

20261012 Quiz

for (MATH210) Applied Complex Variables

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Exercise 36.6. Show that if $z \neq 0$ and a is a real number, then

$$|z^a| = \exp(a \ln |z|) = |z|^a,$$

where the principal value of $|z|^a$ is to be taken.

Solution. For $z \neq 0$, every value of $\log z$ has the form

$$\ln |z| + i(\operatorname{Arg} z + 2k\pi), \quad k \in \mathbb{Z}.$$

Since $a \in \mathbb{R}$, each value of z^a is

$$\exp(a \log z) = \exp(a \ln |z|) \exp(ia(\operatorname{Arg} z + 2k\pi)).$$

The second factor has modulus 1, so every value of z^a has modulus $\exp(a \ln |z|)$. Since $|z| > 0$, its ordinary real power is

$$|z|^a = \exp(a \ln |z|).$$

Thus

$$\boxed{|z^a| = \exp(a \ln |z|) = |z|^a}.$$

□

Exercise 36.8. Let c, c_1, c_2, z be complex numbers with $z \neq 0$. Prove that, if all the powers involved are principal values, then

- (a) $z^{c_1} z^{c_2} = z^{c_1+c_2}$;
- (b) $\frac{z^{c_1}}{z^{c_2}} = z^{c_1-c_2}$;
- (c) $(z^c)^n = z^{cn}$, where $n = 1, 2, \dots$

Solution. Use the principal-value convention $z^w = \exp(w \operatorname{Log} z)$ throughout this exercise.

- (a) By the exponential addition law,

$$z^{c_1} z^{c_2} = e^{c_1 \operatorname{Log} z} e^{c_2 \operatorname{Log} z} = e^{(c_1+c_2) \operatorname{Log} z} = \boxed{z^{c_1+c_2}}.$$

- (b) The denominator is nonzero; hence

$$\frac{z^{c_1}}{z^{c_2}} = \frac{e^{c_1 \operatorname{Log} z}}{e^{c_2 \operatorname{Log} z}} = e^{(c_1-c_2) \operatorname{Log} z} = \boxed{z^{c_1-c_2}}.$$

- (c) Since n is a positive integer, raising a complex number to its n th power has its ordinary single-valued meaning. Thus

$$(z^c)^n = (e^{c \operatorname{Log} z})^n = e^{nc \operatorname{Log} z} = \boxed{z^{cn}}.$$

The common choice of $\operatorname{Log} z$ is essential to these identities; arbitrary independent choices of the multivalued powers need not satisfy them. □

Exercise 36.9. Assuming that $f'(z)$ exists, state the formula for the derivative of $c^{f(z)}$.

Solution. For a nonzero constant c , fix a value L of $\log c$. By Sec. 35,

$$c^{f(z)} = \exp(f(z)L).$$

The chain rule gives

$$\boxed{\frac{d}{dz} c^{f(z)} = c^{f(z)} f'(z) L = c^{f(z)} f'(z) \log c},$$

where the last expression uses the same fixed value of $\log c$ on both sides. In particular, principal values correspond to $L = \operatorname{Log} c$. □

Exercise 38.1. Give details in the derivation of expressions (2), Sec. 37, for the derivatives of $\sin z$ and $\cos z$.

Solution. By their definitions in Sec. 37,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}.$$

Applying the exponential differentiation rule and the chain rule,

$$\begin{aligned} \frac{d}{dz} \sin z &= \frac{ie^{iz} + ie^{-iz}}{2i} = \frac{e^{iz} + e^{-iz}}{2} = \boxed{\cos z}, \\ \frac{d}{dz} \cos z &= \frac{ie^{iz} - ie^{-iz}}{2} = -\frac{e^{iz} - e^{-iz}}{2i} = \boxed{-\sin z}. \end{aligned}$$

□

Exercise 38.4. Verify the identity $\sin^2 z + \cos^2 z = 1$ using

- (a) identity (6) and relations (3) in Sec. 37;
- (b) the lemma in Sec. 28 and the fact that the entire function $f(z) = \sin^2 z + \cos^2 z - 1$ has zero values along the real axis.

Solution.

- (a) Setting $z_1 = z$ and $z_2 = -z$ in the addition formula

$$\cos(z_1 + z_2) = \cos z_1 \cos z_2 - \sin z_1 \sin z_2$$

and using $\cos(-z) = \cos z$, $\sin(-z) = -\sin z$, we get

$$1 = \cos 0 = \cos^2 z + \sin^2 z.$$

- (b) The function $f(z) = \sin^2 z + \cos^2 z - 1$ is entire. For every real x , the real trigonometric identity gives $f(x) = 0$. Since the real axis contains a sequence of distinct zeros converging to, for example, 0, the lemma in Sec. 28 implies $f \equiv 0$ throughout $\mathbb{C}$. Thus $\sin^2 z + \cos^2 z = 1$ for all $z \in \mathbb{C}$. □

Exercise 38.15. Find all roots of $\sin z = \cosh 4$ by equating the real parts and then the imaginary parts of $\sin z$ and $\cosh 4$.

Solution. Write $z = x + iy$, with $x, y \in \mathbb{R}$. Formula (13), Sec. 37, gives

$$\sin z = \sin x \cosh y + i \cos x \sinh y.$$

Consequently,

$$\sin x \cosh y = \cosh 4, \quad \cos x \sinh y = 0.$$

If $y = 0$, the first equation becomes $\sin x = \cosh 4 > 1$, which is impossible. Hence $\cos x = 0$, so $x = \pi/2 + n\pi$. The first equation now reads $(-1)^n \cosh y = \cosh 4 > 0$, forcing n to be even and $\cosh y = \cosh 4$. Since $\cosh y$ is even and strictly increasing for $y > 0$, $y = \pm 4$. All roots are therefore

$$z = \frac{\pi}{2} + 2n\pi \pm 4i \quad (n \in \mathbb{Z}).$$

□

Exercise 39.1. Verify that the derivatives of $\sinh z$ and $\cosh z$ are as stated in equations (2), Sec. 39.

Solution. From the definitions

$$\sinh z = \frac{e^z - e^{-z}}{2}, \quad \cosh z = \frac{e^z + e^{-z}}{2},$$

we obtain

$$\begin{aligned} \frac{d}{dz} \sinh z &= \frac{e^z + e^{-z}}{2} = \boxed{\cosh z}, \\ \frac{d}{dz} \cosh z &= \frac{e^z - e^{-z}}{2} = \boxed{\sinh z}. \end{aligned}$$

□

Exercise 39.8. Give details showing that the zeros of $\sinh z$ and $\cosh z$ are as in the theorem in Sec. 39.

Solution. The identities from Sec. 39 yield

$$\sinh z = -i \sin(iz), \quad \cosh z = \cos(iz).$$

By the zero sets of sine and cosine in Sec. 38,

$$\sinh z = 0 \iff iz = n\pi \iff z = -n\pi i \quad (n \in \mathbb{Z}),$$

which is exactly the set $z = n\pi i$, since n ranges over all integers. Similarly,

$$\cosh z = 0 \iff iz = \frac{\pi}{2} + n\pi \iff z = -\left(\frac{\pi}{2} + n\pi\right)i.$$

Reindexing with $m = -n - 1$ gives the equivalent form

$$z = \left(\frac{\pi}{2} + m\pi\right)i \quad (m \in \mathbb{Z}).$$

Conversely, the displayed identities show directly that every listed point is a zero, and no others are possible.

□

Exercise 39.16. By using one of the identities (9) and (10) in Sec. 39 and then proceeding as in Exercise 15, Sec. 38, find all roots of

- (a) $\sinh z = i$;
- (b) $\cosh z = \frac{1}{2}$.

Solution. Write $z = x + iy$, where $x, y \in \mathbb{R}$. From Sec. 39,

$$\begin{aligned}\sinh z &= \sinh x \cos y + i \cosh x \sin y, \\ \cosh z &= \cosh x \cos y + i \sinh x \sin y.\end{aligned}$$

- (a) The equation $\sinh z = i$ becomes

$$\sinh x \cos y = 0, \quad \cosh x \sin y = 1.$$

Since $\cosh x \geq 1$, if $\cos y = 0$, then $\sin y = \pm 1$; the second equation requires $\sin y = 1$ and $\cosh x = 1$, hence $x = 0$. If instead $\sinh x = 0$, we again have $x = 0$, and the second equation gives $\sin y = 1$. Thus every root is

$$z = \left(\frac{\pi}{2} + 2n\pi \right) i \quad (n \in \mathbb{Z}).$$

- (b) The equation $\cosh z = 1/2$ gives

$$\cosh x \cos y = \frac{1}{2}, \quad \sinh x \sin y = 0.$$

If $\sin y = 0$, then $\cos y = \pm 1$, and the first equation would require $\cosh x = 1/2$ or $-1/2$, both impossible. Therefore $\sinh x = 0$, i.e. $x = 0$, and $\cos y = 1/2$. It follows that

$$z = \left(2n\pi \pm \frac{\pi}{3} \right) i \quad (n \in \mathbb{Z}).$$

□

Exercise 40.2. Solve the equation $\sin z = 2$ for z by

- (a) equating real parts and then imaginary parts in that equation;
- (b) using expression (2), Sec. 40, for $\sin^{-1} z$.

Solution.

- (a) Let $z = x + iy$. Equating components of $\sin z = \sin x \cosh y + i \cos x \sinh y = 2$ gives

$$\sin x \cosh y = 2, \quad \cos x \sinh y = 0.$$

If $y = 0$, then $\sin x = 2$, which is impossible. Thus $\cos x = 0$, so $x = \pi/2 + n\pi$. Positivity of 2 forces n to be even, and $\cosh y = 2$. Solving $e^y + e^{-y} = 4$ yields

$$y = \pm \ln(2 + \sqrt{3}).$$

- (b) Expression (2), Sec. 40, gives the multivalued inverse formula

$$\sin^{-1} 2 = -i \log(2i + (1 - 4)^{1/2}) = -i \log(i(2 \pm \sqrt{3})).$$

Both $2 \pm \sqrt{3}$ are positive, so all logarithmic values are

$$\ln(2 \pm \sqrt{3}) + i \left(\frac{\pi}{2} + 2n\pi \right), \quad n \in \mathbb{Z}.$$

Multiplication by $-i$, together with $\ln(2 - \sqrt{3}) = -\ln(2 + \sqrt{3})$, gives the same solutions as in part (a).

Therefore

$$z = \frac{\pi}{2} + 2n\pi \pm i \ln(2 + \sqrt{3}) \quad (n \in \mathbb{Z}).$$

□

Exercise 40.3. Solve the equation $\cos z = 2$ for z .

Solution. For $z = x + iy$, use the identity

$$\cos z = \cos x \cosh y - i \sin x \sinh y.$$

The equation $\cos z = 2$ is therefore equivalent to

$$\cos x \cosh y = 2, \quad \sin x \sinh y = 0.$$

The case $y = 0$ is impossible because then $\cos x = 2$. Hence $\sin x = 0$, so $x = n\pi$. As $\cosh y > 0$, the real-part equation forces n to be even and $\cosh y = 2$. Thus

$$z = 2n\pi \pm i \ln(2 + \sqrt{3}) \quad (n \in \mathbb{Z}).$$

□

Exercise 42.2. Evaluate the following integrals:

- (a) $\int_0^1 (1 + it)^2 dt$;
- (b) $\int_1^2 \left(\frac{1}{t} - i\right)^2 dt$;
- (c) $\int_0^{\pi/6} e^{i2t} dt$;
- (d) $\int_0^\infty e^{-zt} dt$, where $\operatorname{Re} z > 0$.

Solution. The integrals of complex-valued functions of a real variable are evaluated componentwise (Sec. 42).

- (a) Expanding the integrand,

$$\int_0^1 (1 + it)^2 dt = \int_0^1 (1 - t^2 + 2it) dt = \boxed{\frac{2}{3} + i}.$$

- (b) Since $(t^{-1} - i)^2 = t^{-2} - 1 - 2i/t$,

$$\begin{aligned} \int_1^2 \left(\frac{1}{t} - i\right)^2 dt &= \left(-\frac{1}{t} - t - 2i \ln t\right) \Big|_1^2 \\ &= \boxed{-\frac{1}{2} - i \ln 4}. \end{aligned}$$

- (c) Using $\frac{d}{dt} e^{2it} = 2ie^{2it}$,

$$\begin{aligned} \int_0^{\pi/6} e^{2it} dt &= \frac{e^{i\pi/3} - 1}{2i} = \frac{\frac{1}{2} + i\frac{\sqrt{3}}{2} - 1}{2i} \\ &= \boxed{\frac{\sqrt{3}}{4} + \frac{i}{4}}. \end{aligned}$$

(d) Write $z = x + iy$ with $x > 0$. Then

$$|e^{-zT}| = e^{-xT} \longrightarrow 0 \quad (T \rightarrow \infty).$$

Since $z \neq 0$, the improper integral converges and

$$\int_0^\infty e^{-zt} dt = \lim_{T \rightarrow \infty} \frac{1 - e^{-zT}}{z} = \boxed{\frac{1}{z}}.$$

□

Exercise 42.3. Show that if m and n are integers, then

$$\int_0^{2\pi} e^{im\theta} e^{-in\theta} d\theta = \begin{cases} 0 & \text{when } m \neq n, \\ 2\pi & \text{when } m = n. \end{cases}$$

Solution. Put $k = m - n \in \mathbb{Z}$. If $k = 0$, the integrand is 1, so the integral is 2π . If $k \neq 0$, then

$$\int_0^{2\pi} e^{ik\theta} d\theta = \frac{e^{2\pi ik} - 1}{ik} = 0,$$

because $e^{2\pi ik} = 1$ for every integer k . This proves both cases.

□

Exercise 43.4. Verify expression (14), Sec. 43, for the derivative of $Z(\tau) = z(\phi(\tau))$.

Suggestion. Write $Z(\tau) = x(\phi(\tau)) + iy(\phi(\tau))$ and apply the chain rule for real-valued functions of a real variable.

Solution. Write $z(t) = x(t) + iy(t)$ and $t = \phi(\tau)$. By the chain rule in real calculus,

$$\begin{aligned} Z'(\tau) &= \frac{d}{d\tau} x(\phi(\tau)) + i \frac{d}{d\tau} y(\phi(\tau)) \\ &= x'(\phi(\tau))\phi'(\tau) + iy'(\phi(\tau))\phi'(\tau) \\ &= (x'(\phi(\tau)) + iy'(\phi(\tau)))\phi'(\tau). \end{aligned}$$

Therefore

$$\boxed{Z'(\tau) = z'(\phi(\tau))\phi'(\tau)},$$

which is expression (14) in Sec. 43.

□

Exercise 43.5. Suppose that a function $f(z)$ is analytic at a point $z_0 = z(t_0)$ lying on a smooth arc $z = z(t)$ ($a \leq t \leq b$). Show that if $w(t) = f(z(t))$, then

$$w'(t) = f'(z(t))z'(t)$$

when $t = t_0$.

Suggestion. Write $f(z) = u(x, y) + iv(x, y)$ and $z(t) = x(t) + iy(t)$, so that

$$w(t) = u(x(t), y(t)) + iv(x(t), y(t)).$$

Then apply the chain rule in calculus for functions of two real variables to write

$$w' = (u_x x' + u_y y') + i(v_x x' + v_y y'),$$

and use the Cauchy–Riemann equations.

Solution. At $t = t_0$, the real-variable chain rule gives

$$\begin{aligned} w'(t) &= (u_x x' + u_y y') + i(v_x x' + v_y y') \\ &= (u_x + i v_x) x' + (u_y + i v_y) y', \end{aligned}$$

where the partial derivatives are evaluated at $(x(t_0), y(t_0))$ and x', y' at t_0 . Since f is analytic at z_0 , the Cauchy–Riemann equations imply

$$u_y = -v_x, \quad v_y = u_x,$$

and therefore

$$u_y + i v_y = -v_x + i u_x = i(u_x + i v_x).$$

Moreover, $f'(z_0) = u_x + i v_x$. It follows that

$$\begin{aligned} w'(t_0) &= f'(z_0)(x'(t_0) + i y'(t_0)) \\ &= \boxed{f'(z(t_0))z'(t_0)}. \end{aligned}$$

□

Exercise 46.3. Let $f(z) = \pi \exp(\pi \bar{z})$, and let C be the boundary of the square with vertices $0, 1, 1 + i, i$, oriented counterclockwise. Evaluate $\int_C f(z) \, dz$.

Solution. Parametrize the four directed sides, in order, by

$$\begin{aligned} C_1 : z &= t, & 0 \leq t \leq 1, \\ C_2 : z &= 1 + it, & 0 \leq t \leq 1, \\ C_3 : z &= 1 - t + i, & 0 \leq t \leq 1, \\ C_4 : z &= i(1 - t), & 0 \leq t \leq 1. \end{aligned}$$

Using the conjugate of each parametrization, the contour-integral definition in Sec. 44 gives

$$\begin{aligned} \int_{C_1} f(z) \, dz &= \pi \int_0^1 e^{\pi t} \, dt = e^\pi - 1, \\ \int_{C_2} f(z) \, dz &= i\pi e^\pi \int_0^1 e^{-i\pi t} \, dt = 2e^\pi, \\ \int_{C_3} f(z) \, dz &= \pi \int_0^1 e^{\pi(1-t)} \, dt = e^\pi - 1, \\ \int_{C_4} f(z) \, dz &= -i\pi \int_0^1 e^{-i\pi(1-t)} \, dt = -2. \end{aligned}$$

Adding the four contributions,

$$\boxed{\int_C \pi e^{\pi \bar{z}} \, dz = 4(e^\pi - 1)}.$$

□

Exercise 46.9. Let C denote the positively oriented unit circle $|z| = 1$ about the origin.

(a) Show that if $f(z)$ is the principal branch

$$z^{-3/4} = \exp\left(-\frac{3}{4} \operatorname{Log} z\right) \quad (|z| > 0, -\pi < \operatorname{Arg} z < \pi)$$

of $z^{-3/4}$, then

$$\int_C f(z) \, dz = 4\sqrt{2}i.$$

(b) Show that if $g(z)$ is the branch

$$z^{-3/4} = \exp\left(-\frac{3}{4} \log z\right) \quad (|z| > 0, 0 < \arg z < 2\pi)$$

of the same power function, then

$$\int_C g(z) \, dz = -4 + 4i.$$

This exercise demonstrates how the value of an integral of a power function depends in general on the branch that is used.

Solution. We evaluate the contour integrals using arguments in the intervals defining the respective branches. At a point where the circle meets a branch cut, the corresponding integral is interpreted by the one-sided limits of the parametrized integrals; isolated endpoint values do not affect those limits.

(a) Take $z = e^{i\theta}$ with $-\pi < \theta < \pi$. Then $\text{Log } z = i\theta$, $dz = ie^{i\theta} \, d\theta$, and

$$\begin{aligned} \int_C f(z) \, dz &= \int_{-\pi}^{\pi} e^{-3i\theta/4} ie^{i\theta} \, d\theta \\ &= \int_{-\pi}^{\pi} ie^{i\theta/4} \, d\theta \\ &= 4(e^{i\pi/4} - e^{-i\pi/4}) = \boxed{4\sqrt{2}i}. \end{aligned}$$

(b) Let $h(z) = \ln |z| + i\theta$, where $0 < \theta < 2\pi$, denote the single-valued logarithmic branch written as $\log z$ in the exercise. Parametrize C by $z = e^{i\theta}$, $0 < \theta < 2\pi$. Then $h(z) = i\theta$, so

$$\begin{aligned} \int_C g(z) \, dz &= \int_0^{2\pi} e^{-3i\theta/4} ie^{i\theta} \, d\theta \\ &= 4(e^{i\pi/2} - 1) = \boxed{-4 + 4i}. \end{aligned}$$

The difference results from the distinct argument intervals used for the two branches.

□

Exercise 46.10. With the aid of the result in Exercise 3, Sec. 42, evaluate

$$\int_C z^m \bar{z}^n \, dz,$$

where $m, n \in \mathbb{Z}$ and C is the unit circle $|z| = 1$, oriented counterclockwise.

Solution. Use the parametrization $z = e^{i\theta}$, $0 \leq \theta \leq 2\pi$. On C ,

$$\bar{z} = e^{-i\theta}, \quad dz = ie^{i\theta} \, d\theta.$$

Thus

$$\begin{aligned} \int_C z^m \bar{z}^n \, dz &= i \int_0^{2\pi} e^{im\theta} e^{-in\theta} e^{i\theta} \, d\theta \\ &= i \int_0^{2\pi} e^{i(m+1)\theta} e^{-in\theta} \, d\theta. \end{aligned}$$

By Exercise 42.3, the last integral is 2π when $m+1 = n$ and 0 otherwise. Hence

$$\boxed{\int_C z^m \bar{z}^n \, dz = \begin{cases} 2\pi i & \text{if } n = m + 1, \\ 0 & \text{if } n \neq m + 1. \end{cases}}$$

□